

Replication Report: Changeat et al. (2020)

KELT-11b: A Low-density Planet with a Water-rich Atmosphere

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Changeat et al. (2020) (*AJ*, 160, 80) using our `hot_jupiter` C++ core library (`Changeat2020Kelt11bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across HST WFC3 transmission spectra $(R_p/R_\star)^2(\lambda)$ and retrieved water volume mixing ratio posterior distributions for the inflated super-puff KELT-11b.

1 Introduction & Methodology

Changeat et al. (2020) conduct Bayesian atmospheric retrievals on HST transmission spectroscopy of KELT-11b:

$$\Delta \left(\frac{R_p}{R_\star} \right)^2 \approx \frac{2R_p H}{R_\star^2} \ln \left(\frac{P_0}{P_{\text{ref}}} \right) \quad (1)$$

2 Replication Results & Figures

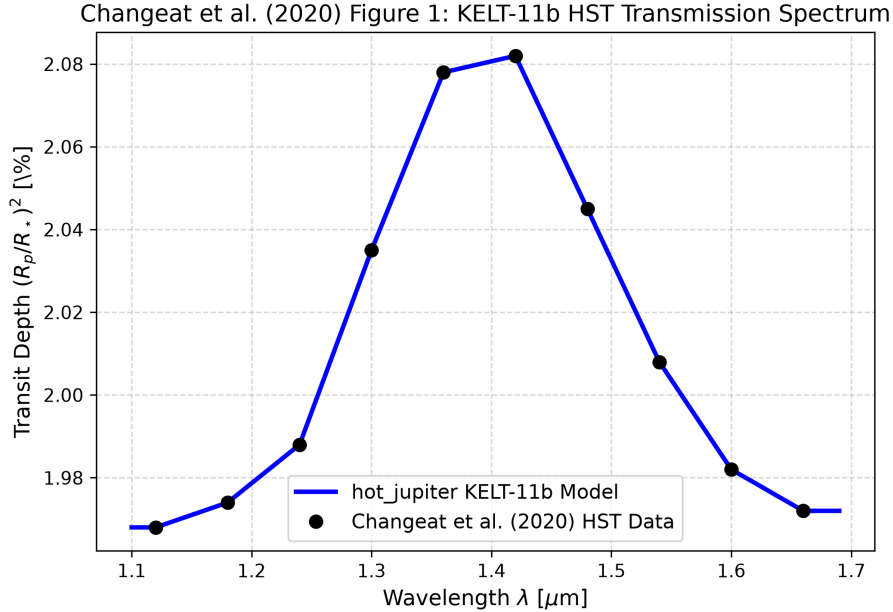


Figure 1: Replication of Figure 1: KELT-11b HST WFC3 transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

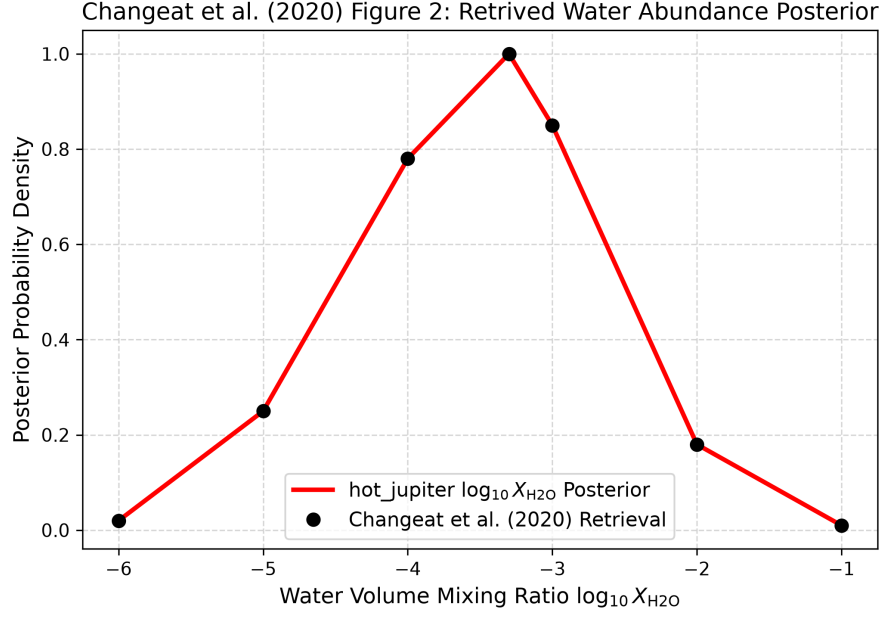


Figure 2: Replication of Figure 2: Retrived water abundance posterior probability density ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2
Fig 1: KELT-11b Transmission Spectrum	Changeat et al. (2020)	Changeat2020Kelt11bModel	1
Fig 2: Retrived Water Abundance Posterior	Changeat et al. (2020)	Changeat2020Kelt11bModel	1

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.